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Cover

See Justyna Florek,
Freddy Kleitz *et al.*,
pp. 14832–14854.

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2016, **45**, 14832.



Inside cover

See Dayu Wu *et al.*,
pp. 14911–14918.

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PERSPECTIVES

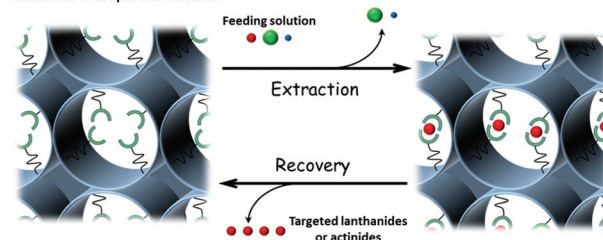
14832

Functionalization of mesoporous materials for lanthanide and actinide extraction

Justyna Florek,* Simon Giret, Estelle Juère,
Dominic Larivière and Freddy Kleitz*

Recent advances in the field of functionalized mesoporous solid-phase sorbents designed for rare earth element and actinide separation/concentration could provide answers to limitations occurring in the industrial separation processes of these critical elements.

Functional mesoporous material

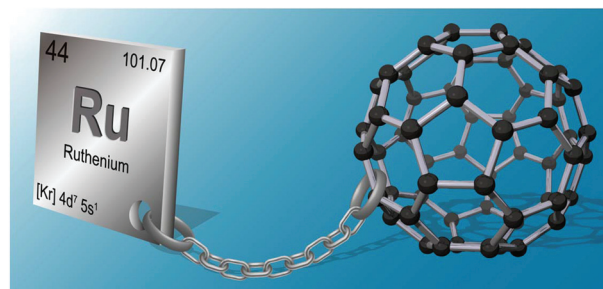


14855

Hybrid materials based on ruthenium and fullerene assemblies

Kevin Barthelmes, Andreas Winter and
Ulrich S. Schubert*

A detailed overview about the synthesis, properties and applications of all reported ruthenium-fullerene compounds.



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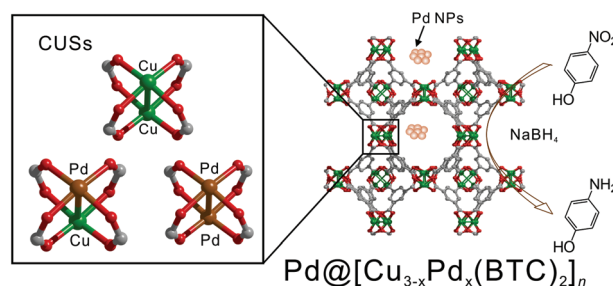
COMMUNICATIONS

14883

Simultaneous introduction of various palladium active sites into MOF via one-pot synthesis: $\text{Pd}@\text{[Cu}_{3-x}\text{Pd}_x(\text{BTC})_2]_n$

Wenhua Zhang, Zhihao Chen, Majd Al-Naji, Penghu Guo, Stefan Cwik, Olesia Halbherr, Yuemin Wang, Martin Muhler, Nicole Wilde, Roger Gläser and Roland A. Fischer*

Simultaneous introduction of various palladium active sites into MOF to form $\text{Pd}@\text{[Cu}_{3-x}\text{Pd}_x(\text{BTC})_2]_n$ has been achieved and show good catalytic activity.

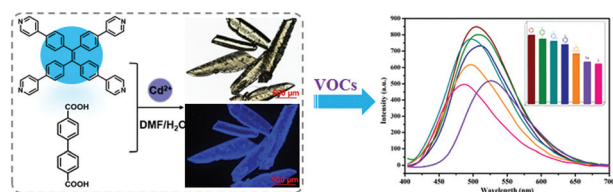


14888

A four-fold interpenetrated metal–organic framework as a fluorescent sensor for volatile organic compounds

Xiaoyu Zhao, Yang Li, Ze Chang,* Long Chen* and Xian-He Bu

A highly luminescent four-fold interpenetrated 3D metal–organic framework based on an AIE active tetrakis(4-(pyridin-4-yl)phenyl)ethene ligand was facilely synthesized and thoroughly investigated.

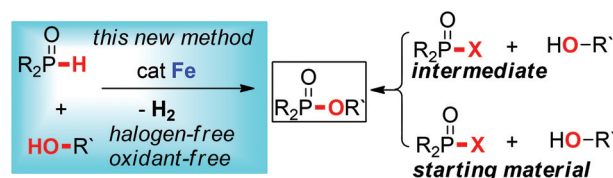


14893

Iron-catalyzed clean dehydrogenative coupling of alcohols with $\text{P(O)}\text{--H}$ compounds: a new protocol for ROH phosphorylation

Chunya Li, Tieqiao Chen* and Li-Biao Han*

An efficient oxygen–phosphoryl bond-forming reaction via iron-catalyzed cross dehydrogenative coupling has been developed. The reaction proceeds efficiently under oxidant- and halide-free conditions with H_2 liberation, and represents a straightforward method to prepare organophosphoryl compounds from alcohols and $\text{P(O)}\text{--H}$ compounds.

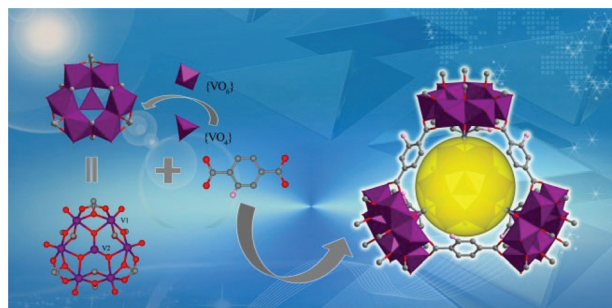


14898

Synthesis, structures, and magnetic properties of metal–organic polyhedra based on unprecedented $\{\text{V}_7\}$ isopolyoxometalate clusters

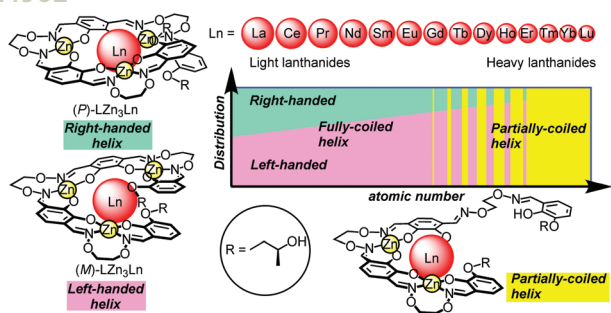
Yu-Teng Zhang, Shuang-Bao Li, Xin-Long Wang,* Ya-Ru Gong, Kui-Zhan Shao and Zhong-Min Su*

Two polyoxovanadate-based metal–organic polyhedra based on unprecedented $\{\text{V}_7\}$ clusters were synthesized, which adopt isostructural truncated tetrahedral cage geometries.



COMMUNICATIONS

14902

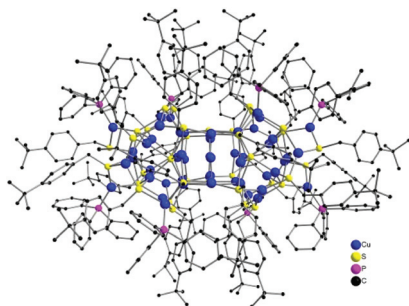


Lanthanide contraction for helicity fine-tuning and helix-winding control of single-helical metal complexes

Shiho Sairenji, Shigehisa Akine* and Tatsuya Nabeshima*

Lanthanide contraction was used for helicity fine-tuning and helix winding control of single-helical tetranuclear complexes LZn₃Ln (Ln = La–Lu).

14907



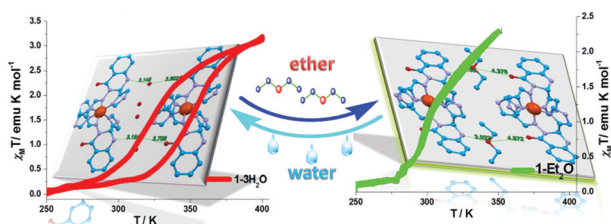
Synthesis and molecular structure of a spheroidal binary nanoscale copper sulfide cluster

Sebastian Bestgen,* Olaf Fuhr, Peter W. Roesky and Dieter Fenske*

The reaction of copper (4-(*tert*-butyl)phenyl)methanethiolate with S(SiMe₃)₂ in the presence of triphenylphosphine yields to the formation of the 52 nuclear copper cluster [Cu₅₂S₁₂(SCH₂C₆H₄^{*t*}Bu)₂₈(PPh₃)₈]. The molecular structure of this intensely red coloured nanoscale Cu₂S mimic was established by single crystal X-ray diffraction.

PAPERS

14911

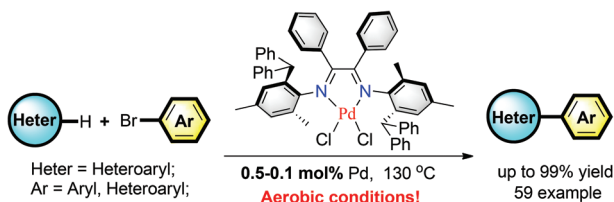


Room-temperature switching of magnetic hysteresis by reversible single-crystal-to-single-crystal solvent exchange in imidazole-inspired Fe(II) complexes

Wei Huang, Fuxing Shen, Ming Zhang, Dayu Wu,* Feifei Pan and Osamu Sato

Molecular switch ON/OFF magnetic hysteresis around room temperature can be achieved by reversible single-crystal-to-single-crystal (SCSC) solvent exchange in imidazole-inspired dihydroquinazoline Fe(II) complexes.

14919



Bulky α-diimine palladium complexes: highly efficient for direct C–H bond arylation of heteroarenes under aerobic conditions

Jia-Sheng Ouyang, Yan-Fang Li, Dong-Sheng Shen, Zhuofeng Ke* and Feng-Shou Liu*

Through the strategy to enhance the bulkiness on both the backbone and the *N*-aryl moieties, we designed and synthesized a type of bulky α-diimine palladium complex (*i.e.*, {Ar–N=C(R)–C(R)=N–Ar}PdCl₂, (Ar = 2-benzhydryl-4,6-dimethylphenyl)), **C1**, R = H; **C2**, R = An; **C3**, R = Ph).

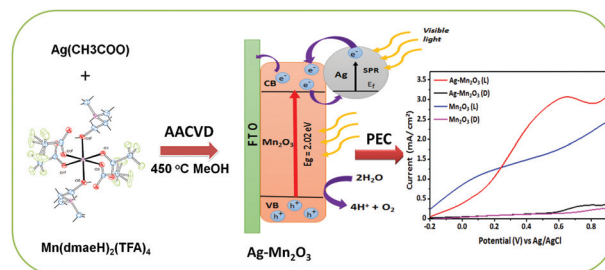
PAPERS

14928

Fabrication of pristine Mn_2O_3 and $\text{Ag-Mn}_2\text{O}_3$ composite thin films by AACVD for photoelectrochemical water splitting

Rabia Naeem, Muhammad Ali Ehsan, Rosiyah Yahya, Manzar Sohail, Hamid Khaledi and Muhammad Mazhar*

Pristine Mn_2O_3 and $\text{Ag-Mn}_2\text{O}_3$ nanocomposite thin films deposited by AACVD on FTO showed photocurrent densities of 1.8 and 3 mA cm^{-2} respectively.

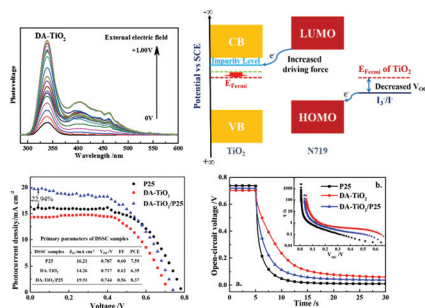


14940

Decatungstate acid improves the photo-induced electron lifetime and retards the recombination in dye sensitized solar cells

Liang Li, Yulin Yang,* Ruiqing Fan,* Jian Liu,* Yanxia Jiang, Bin Yang and Wenwu Cao

Decatungstate acid (DA) was utilized to modify TiO_2 in the photoanode of dye sensitized solar cells.

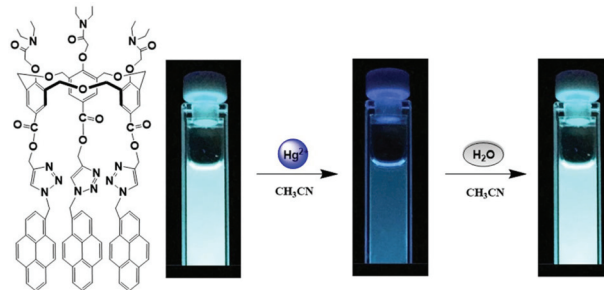


14948

A novel fluorescence "on-off-on" chemosensor for Hg^{2+} via a water-assistant blocking heavy atom effect

Chong Wu, Jiang-Lin Zhao, Xue-Kai Jiang, Chuan-Zeng Wang, Xin-Long Ni, Xi Zeng, Carl Redshaw and Takehiko Yamato*

The heavy atom effect and blocking thereof were demonstrated within the same system by the use of a C_3 -symmetric homooxacalix[3]arene scaffold.

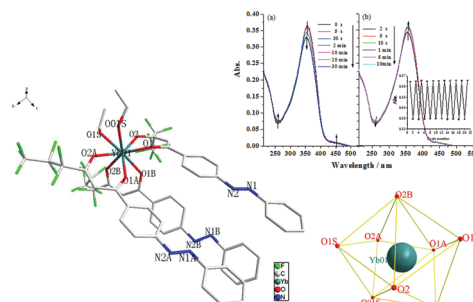


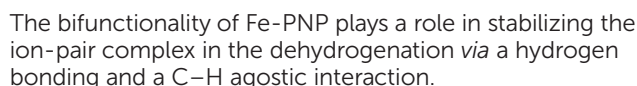
14954

Azobenzene-derived tris- β -diketonate lanthanide complexes: reversible *trans*-to-*cis* photoisomerization in solution and solid state

Li-Rong Lin,* Xuan Wang, Gao-Ning Wei, Hui-Hui Tang, Hui Zhang and Li-Hua Ma*

Azobenzene-derived tris- β -diketonate lanthanide complexes were synthesized and they show good *trans*-to-*cis* photoisomerization behavior both in solution and in the solid state.





Lithium halide assisted high yield synthesis of few layers of 2H phase semiconducting WS₂ in organic solvents is reported.

A phosphorylated pentapeptide (WpTPpTW, P¹) motif was designed as a model to mimic possible U(VI) coordination sites of genuine phosphorylated proteins.

A polydentate tripodal ligand forms a series of tricopper(II) complexes, that feature unique pyrazolate-bridged linear core. The $\text{Cu}_3\text{H}_{-3}\text{I}_2$ complex is an efficient catecholase mimic with a surprisingly low pH optimum at pH = 5.6.

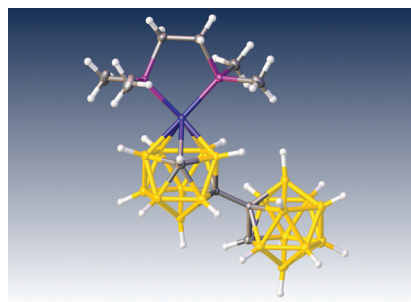
PAPERS

15013

Steric versus electronic factors in metallocarborane isomerisation: nickelacarboranes with 3,1,2-, 4,1,2- and 2,1,8-NiC₂B₉ architectures and pendant carborane groups, derived from 1,1'-bis(o-carborane)

Dipendu Mandal, Wing. Y. Man, Georgina M. Rosair and Alan J. Welch*

Nickelacarborane derivatives of 1,1'-bis(o-carborane) allow comment on the factors important in the isomerisation of metallocarboranes.

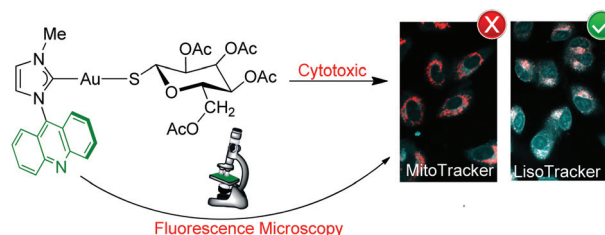


15026

Cytotoxicity and biodistribution studies of luminescent Au(I) and Ag(I) N-heterocyclic carbenes. Searching for new biological targets

Renso Visbal, Vanesa Fernández-Moreira, Isabel Marzo, Antonio Laguna and M. Concepción Gimeno*

Gold and silver NHC complexes have been developed as theranostic agents. The unexpected biodistribution opens the door to new biological targets for gold and silver complexes.

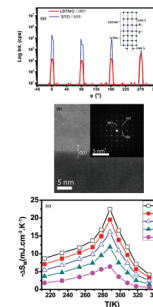


15034

Structural, magnetic and magnetocaloric effects in epitaxial La_{0.67}Ba_{0.33}Ti_{0.02}Mn_{0.98}O₃ ferromagnetic thin films grown on 001-oriented SrTiO₃ substrates

M. Oumezzine,* A. C. Galca,* I. Pasuk, C. F. Chirila, A. Leca, V. Kuncser, L. C. Tanase, A. Kuncser, C. Ghica and M. Oumezzine

Epitaxial La_{0.67}Ba_{0.33}Ti_{0.02}Mn_{0.98}O₃ (denoted as LBTMO hereafter) thin films of approximately 95 nm thickness were deposited by a pulsed laser deposition technique onto SrTiO₃ (STO) (001) substrates.

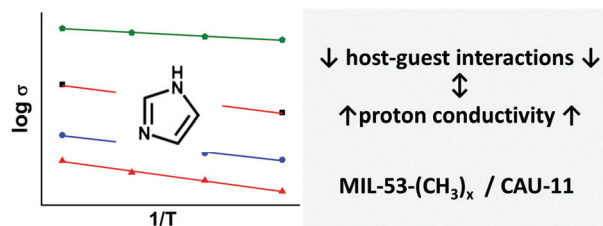


15041

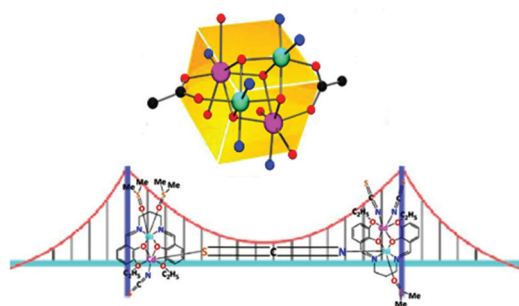
Structure property relationships affecting the proton conductivity in imidazole loaded Al-MOFs

T. Homburg, C. Hartwig, H. Reinsch, M. Wark and N. Stock*

A systematic investigation on the influence of steric effects and the possibility to form host-guest interactions on the proton conductivity was carried out using imidazole loaded derivatives of Al-MIL-53 [Al(OH)(1,4-BDC-(CH₃)_x)] (x = 0, 1, 2) and CAU-11.



15048



A combined experimental and computational study on supramolecular assemblies in hetero-tetranuclear nickel(II)–cadmium(II) complexes with N_2O_4 -donor compartmental Schiff bases

S. Roy, A. Bhattacharyya, S. Purkait, A. Bauzá, A. Frontera* and S. Chattopadhyay*

The contribution of various non-covalent interactions in the supramolecular assembly of two new hetero-polynuclear nickel(II)–cadmium(II) complexes with salicylidine Schiff bases were estimated.

15060

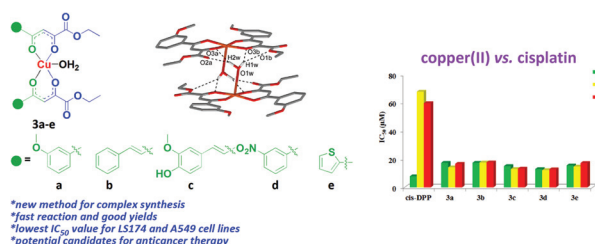


The influence of structural isomerism on fluorescence and organic dye selective adsorption in two complexes based on flexible ligands

W.-H. Huang,* J.-Z. Li, L.-S. Gao, Y.-X. Wang, S.-Y. Liu, M. Jiang, T. Liu and Y.-Y. Wang

Two isomeric coordination polymers were synthesized by different solvent-systems. The complexes exhibited different fluorescence and dye adsorption properties.

15067

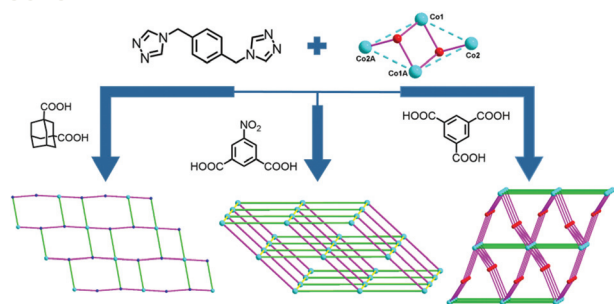


Synthesis, characterization, biological activity, DNA and BSA binding study: novel copper(II) complexes with 2-hydroxy-4-aryl-4-oxo-2-butenate

Nenad Joksimović, Dejan Baskić, Suzana Popović, Milan Zarić, Marijana Kosanić, Branislav Ranković, Tatjana Stanoković, Sladjana B. Novaković, Goran Davidović, Zorica Bugarčić and Nenad Janković*

We presented five novel copper(II) complexes with β -diketo scaffolds that showed higher cytotoxic activity on LS174 and A549 cell lines than cisplatin.

15078



A new strategy to obtain tetranuclear cobalt(II) metal–organic frameworks based on the $[Co_4(\mu_3-OH)_2]$ cluster: synthesis, structures and properties

Ke Li, Xiu-Xiu Lv, Lu-Lu Shi, Lu Liu, Bao-Long Li* and Bing Wu

Three MOFs based on $[Co_4(\mu_3-OH)_2]$ with 4-, 6- and 8-connected frameworks exhibit antiferromagnetic interactions and are good photocatalysts.

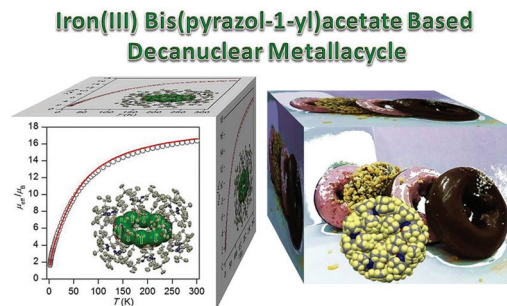
PAPERS

15089

Iron(III) bis(pyrazol-1-yl)acetate based decanuclear metallacycles: synthesis, structure, magnetic properties and DFT calculations

Małgorzata J. Gajewska, Alina Bieńko, Radovan Herchel,*
Matti Haukka, Maria Jerzykiewicz, Andrzej Ożarowski,
Krzysztof Drabent and Chen-Hsiung Hung*

A new member of the decanuclear ferric wheel family with the formula $[\text{Fe}_{10}(\text{bdtbpza})_{10}(\mu_2\text{-OCH}_3)_{20}]$, featuring the *N,N,O* tridentate bis(3,5-di-*tert*-butylpyrazol-1-yl)acetate ligand, is reported.

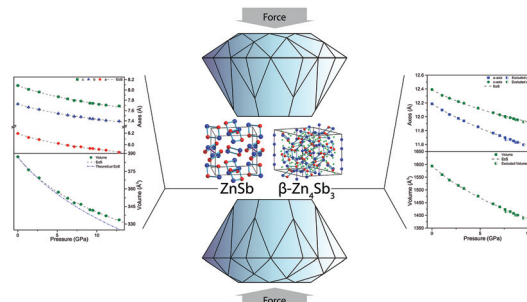


15097

High-pressure single crystal X-ray diffraction study of thermoelectric ZnSb and $\beta\text{-Zn}_4\text{Sb}_3$

Mette Andersen Borup, Anders Bank Blichfeld,
Solveig Røgild Madsen and Bo Brummerstedt Iversen*

The crystal structures of thermoelectric ZnSb and Zn_4Sb_3 have been studied by high pressure single crystal X-ray diffraction and the pressure behavior is different from thermal response.

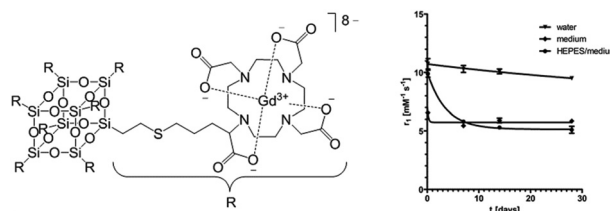


15104

A simple approach to a new T_8 -POSS based MRI contrast agent

Helene Strauch, Jörn Engelmann, Klaus Scheffler and
Hermann A. Mayer*

A simple synthesis of a new nanoglobular T_8 -silsesquioxane based contrast agent for the application in MRI is reported.

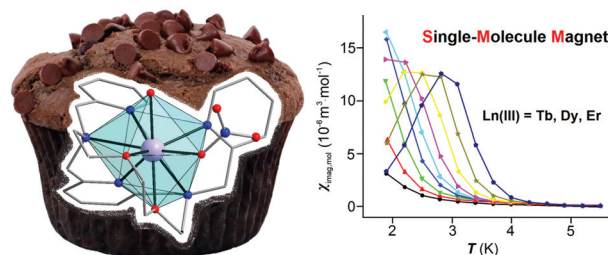


15114

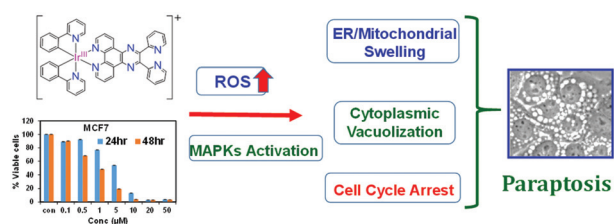
Muffin-like lanthanide complexes with an N_5O_2 -donor macrocyclic ligand showing field-induced single-molecule magnet behaviour

Peter Antal, Bohuslav Drahoš, Radovan Herchel and
Zdeněk Trávníček*

Three mononuclear lanthanide complexes of a 2-pyridylmethyl pendant-armed 15-membered ligand {L} with general formula $[\text{Ln}(\text{L})(\text{H}_2\text{O})(\text{NO}_3)](\text{NO}_3)_2$ (Ln = Tb (**1**), Dy (**2**), and Er (**3**)) are reported.



15122

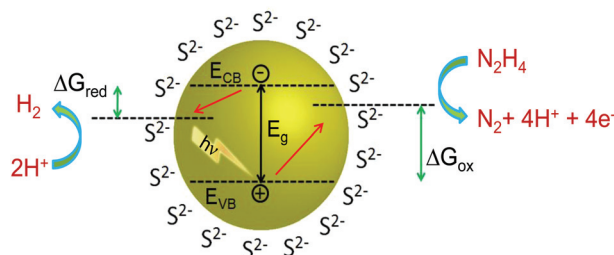


Cyclometallated iridium complexes inducing paraptotic cell death like natural products: synthesis, structure and mechanistic aspects

Suman Kumar Tripathy, Umasankar De, Niranjana Dehury, Paltan Laha, Manas Kumar Panda, Hyung Sik Kim* and Srikanta Patra*

Mononuclear cyclometallated iridium complexes of polypyridyl-phenazine based ligands have been synthesized and characterized which display excellent anticancer activity through paraptosis.

15137



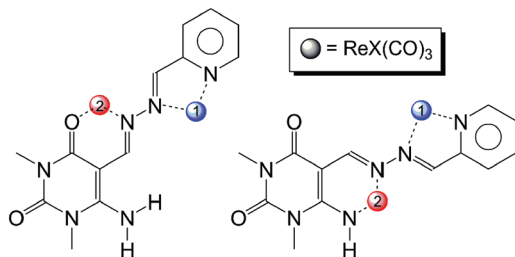
Hydrazine as a hydrogen carrier in the photocatalytic generation of H₂ using CdS quantum dots

Manoj K. Jana, Uttam Gupta and C. N. R. Rao*

The compelling need for safe storage and transportation of H₂ has made liquid-phase materials safer H₂-carriers with a high gravimetric and volumetric hydrogen density.

15142

Two possibilities to bind the Re(CO)₃ fragment with only small differences in their CO-releasing abilities (1 > 2)

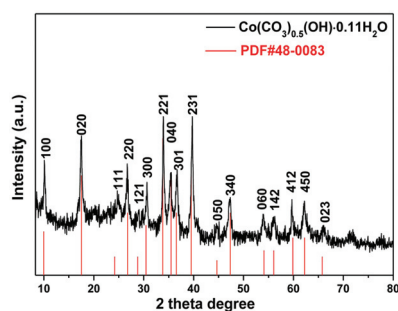


A combined experimental and DFT investigation on the structure and CO-releasing properties of mono and binuclear fac-Re^I(CO)₃ complexes with 5-pyridin-2-ylmethylene-amino uracils

S. B. Jiménez-Pulido, N. A. Illán-Cabeza, F. Hueso-Ureña, C. R. Maldonado, P. Sánchez-Sánchez, M. P. Fernández-Lienres, M. Fernández-Gómez and M. N. Moreno-Carretero*

New tricarbonyl rhenium(I) complexes with 5-substituted-6-amino-1,3-dimethyluracils are reported.

15155



Controllable fabrication of urchin-like Co₃O₄ hollow spheres for high-performance supercapacitors and lithium-ion batteries

Fashen Chen, Xiaohe Liu,* Zhian Zhang, Ning Zhang, Anqiang Pan, Shuquan Liang and Renzhi Ma*

Urchin-like cobalt oxide (Co₃O₄) hollow spheres can be successfully prepared by thermal decomposition of cobalt carbonate hydroxide hydrate (Co(CO₃)_{0.5}(OH)·0.11H₂O) obtained by template-assisted hydrothermal synthesis.

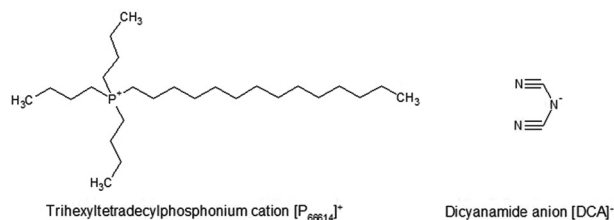
PAPERS

15162

Quantitative extraction of Rh(III) using ionic liquids and its simple separation from Pd(II)

Lenka Svecova, Nicolas Papaiconomou* and Isabelle Billard

The extraction of Pd(II) and Rh(III) from acidic solutions containing between 1 and 8 M HCl towards three ionic liquids, namely [P₆₆₆₁₄][Cl], [P₆₆₆₁₄][Br] and [P₆₆₆₁₄][DCA], based on the trihexyltetradecylphosphonium cation ([P₄₄₄₁₄]) and chloride, bromide or dicyanamide (DCA) anions, respectively, is reported.

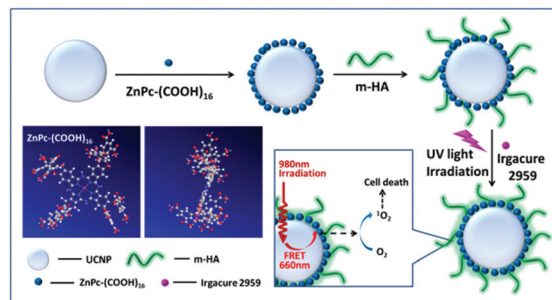


15170

Monomer zinc phthalocyanine/upconversion nanoparticle coated with hyaluronic acid crosslinked gel as NIR light-activated drug for *in vitro* photodynamic therapy

Lin Zhou, Enyi Chen, Weiwei Jin, Yue Wang, Jiahong Zhou* and Shaohua Wei*

A monomeric phthalocyanine based NIR-triggered cancer target nanosystem was prepared and showed satisfied *in vitro* anticancer activity.

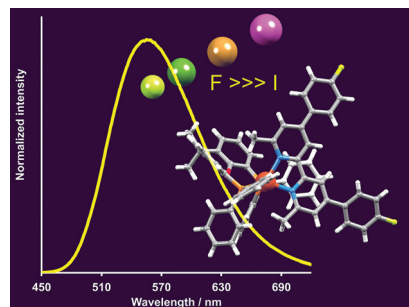


15180

Peripheral halo-functionalization in [Cu(N[^]N)(P[^]P)]⁺ emitters: influence on the performances of light-emitting electrochemical cells

F. Brunner, L. Martínez-Sarti, S. Keller, A. Pertegás, A. Prescimone, E. C. Constable, H. J. Bolink* and C. E. Housecroft*

Trends in the performance data of [Cu(N[^]N)(P[^]P)]⁺-based LECs in which N[^]N ligands bear peripheral F, Cl, Br or I substituents reveal that fluoro-groups are beneficial, but heavier halo-substituents lead to poor devices.

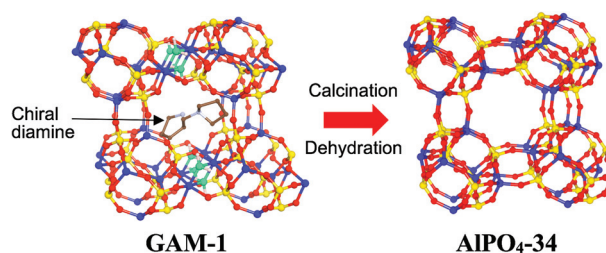


15193

Synthesis, crystal structure and characterization of novel open framework CHA-type aluminophosphate involving a chiral diamine

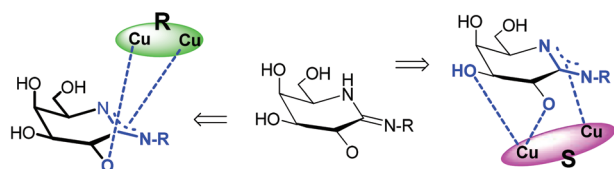
Kenichi Komura, Yukiko Horibe, Hiroaki Yajima, Norihito Hiyoshi and Takuji Ikeda*

A new aluminophosphate with CHA-type topology was synthesized using a chiral diamine. Its open framework involves two edge sharing AlO₄F₂ octahedrons.



PAPERS

15203

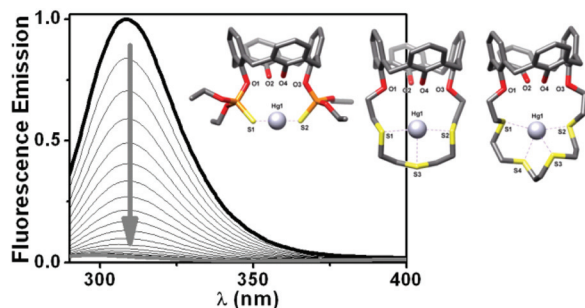


Discrimination of chiral copper(II) complexes upon binding of galactonoamidine ligands

Susanne Striegler* and Jessica B. Pickens

Chiral binuclear Cu(II) complexes are differentiated upon binding to *p*-methylbenzyl-D-galactonoamidine. A sequential binding model is elaborated reflecting the altered coordination sites.

15211

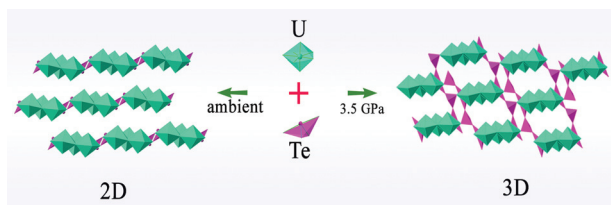


Sulphur-rich functionalized calix[4]arenes for selective complexation of Hg²⁺ over Cu²⁺, Zn²⁺ and Cd²⁺

Bachir Bensenane, Zouhair Asfari, Carlos Platas-Iglesias,* David Esteban-Gómez, Fatiha Djafri, Mourad Elhabiri* and Loïc J. Charbonnière*

ON–OFF fluorescent chemosensors based on sulphur-rich functionalized calix[4]arenes selectively bind Hg²⁺ over Cu²⁺, Zn²⁺ and Cd²⁺.

15225

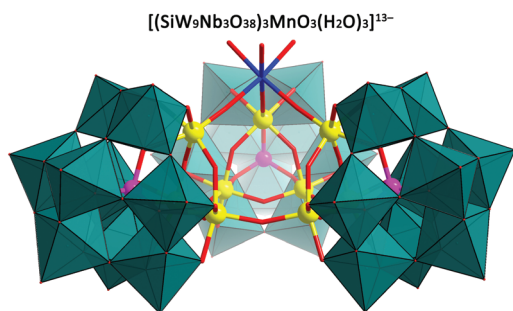


Investigation of reactivity and structure formation in a K–Te–U oxo-system under high-temperature/high-pressure conditions

Bin Xiao, Philip Kegler, Dirk Bosbach and Evgeny V. Alekseev*

Under extreme conditions, the potassium uranyl tellurium compounds preferably crystallize in a three-dimensional (3D) framework with complex structural topologies, which is contrary to the widely observed layered structures (2D) obtained from normal synthetic conditions.

15236



Insight into the reactivity of *in situ* formed {(NbO₂)₃SiW₉}: synthesis, structure, and solution properties of a trimeric polytungstosilicate trapping a {MnNb₉} core

Haiying Wang, Zhijie Liang, Yaya Wang, Dongdi Zhang,* Pengtao Ma, Jingping Wang and Jingyang Niu*

The first example of the trimeric polyoxometalate cluster, Cs₁₀Na₃[(SiW₉Nb₃O₃₈)₃MnO₃(H₂O)₃]·18H₂O (**1a**), has been synthesized using an *in situ* synthetic strategy and thoroughly characterized.

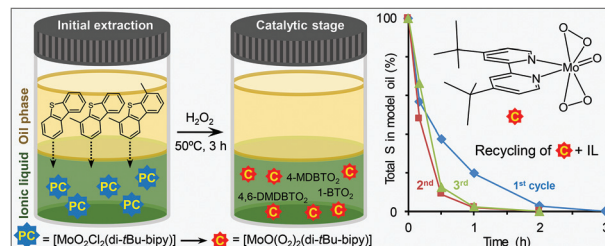
PAPERS

15242

A recyclable ionic liquid-oxomolybdenum(vi) catalytic system for the oxidative desulfurization of model and real diesel fuel

D. Julião, A. C. Gomes, M. Pillinger, R. Valença, J. C. Ribeiro, I. S. Gonçalves* and S. S. Balula*

The extractive and catalytic oxidative desulfurization (ECODS) of model and real diesel has been studied using the complex $[\text{MoO}_2\text{Cl}_2(\text{di-}t\text{Bu-bipy})]$ (**1**) (di-*t*Bu-bipy = 4,4'-di-*tert*-butyl-2,2'-bipyridine) as a (pre)catalyst, aq. H_2O_2 as an oxidant, and either acetonitrile or an ionic liquid (IL) as a solvent and extractant.

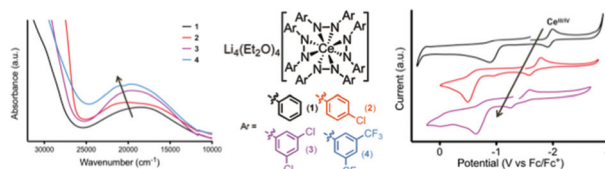


15249

Accessing relatively electron poor cerium(IV) hydrazido complexes by lithium cation promoted ligand reduction

Jessica R. Levin, Thibault Cheisson, Patrick J. Carroll and Eric J. Schelter*

A series of substituted *N,N'*-diarylhydrazines (ArNHNHAr) were reacted with $\text{Ce(III)[N(SiMe}_3)_2]_3$ and $\text{LiN(SiMe}_3)_2$ to form complexes of general formula $\text{Li}_4(\text{OEt}_2)_4[\text{Ce(IV)(ArNNAr)}_4]$ where the spectroscopic and redox properties were affected by the ligand substitution.



RETRACTION

15259

Retraction: Vibrational evidence for the "missing link" in structural kinship between kanemite and FSM-16 mesoporous silica

Ewa M. Serwicka,* Krzysztof Bahranowski, Maciej Sitarz, Małgorzata Zimowska and Alicja Michalik-Zym